

# Daniel Sandi Bratanata

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## PROFILE

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I'm an undergraduate Information Systems student with strong machine learning and deep learning experience. Currently a Research Assistant developing a multimodal emotion-detection system using Kolmogorov–Arnold Networks and LLM embeddings (QWEN, BGE-M3) for a Scopus Q2 publication. Proven technical ability through national competitions, and skilled in end-to-end ML pipelines, from NLP architecture design to deploying models on Hugging Face.

## EXPERIENCES

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### AI Research Assistant - Institut Teknologi Sepuluh Nopember

Aug. 2025 – Present

- Conducted web-scraping to performed preliminary research proposing the use of Kolmogorov–Arnold Network (KAN) and the CBMP mechanism for emotion classification.
- Developed a multimodal model combining text embeddings (Qwen-0.6B, BGE M3, Jina V3) and image embeddings for early detection of adolescent mental-health issues; research targeted for publication in the International Journal of Intelligent and Engineering Systems (Scopus Q2).

### Programming Teacher - Timedoor

Apr. 2025 - Present

- Taught Web Programming (HTML, CSS, JavaScript) to 6+ classes, providing hands-on, industry-relevant training.
- Delivered Artificial Intelligence Fundamentals, including practical image-processing projects such as eye and lip detection using modern AI techniques.

### Teaching Assistant - Institut Teknologi Sepuluh Nopember

Jul. 2025 - Present

- Supported lecturers in designing and delivering hands-on learning sessions on LinkedList, ArrayList, and array implementations for 50+ students, integrating concepts into their final projects
- Managed assignment tracking and coordinated assistant schedules using Microsoft Excel, optimizing workload distribution and ensuring smooth academic operations.

## EDUCATION

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2023 - Present

**Institut Teknologi Sepuluh Nopember**

Bachelor of Information System

**GPA – 3.49**

## ORGANIZATIONS

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### Data Scientist Staff - AVALON AI (PT.Automasi Visi Analitika)

Nov. 2024 – Nov. 2025

- Developed an XGBoost-based machine learning model to predict fraudulent transactions using features such as time, location, and store data, improving detection accuracy.
- Built a CNN-based multiclass image classification model to identify hoodies and t-shirts in 5 colors, enabling efficient automated product recognition.

### Senior Backend Engineer – Information System Expo

Apr. 2025 – Dec. 2025

- Designed database entity models and developed key APIs using the Gonic framework in Golang.
- Architected and optimized a PostgreSQL infrastructure that reliably supported three consecutive high-traffic events, scaling seamlessly to serve 150+ concurrent participants with zero downtime and consistent sub-second query performance.

## Event Staff - Avalon Expo Innovation Competition

Apr. 2025 – Dec. 2025

- Installed and configured Starlink to ensure stable connectivity and seamless technical operations during the event.
- Developed comprehensive Standard Operating Guidelines (Juklak/Juknis), outlining workflows, procedures, and technical instructions for committee members and participants.

## SKILLS & TOOLS

### Hard Skills

- Programming & Data: Python, SQL, Fast API
- Machine Learning & Deep Learning: TensorFlow, Scikit-learn, PyTorch, Sentence Transformer, OpenCV
- Data Visualization: PowerBI, Matplotlib, Seaborn, Plotly
- Data Engineering: PostgreSQL, Pandas, NumPy
- Tools: Anaconda, Visual Studio Code

### Soft Skills

- Communication
- Teamwork
- Problem-Solving
- Adaptability
- Time Management

### Languages

- Indonesia (Native)
- English (Proficient)

## ACHIEVEMENTS

### 3rd Place Data Competition - Multimedia Nusantara University

Oct. 2025

- Analyzed 2020 Open Source Transjakarta dataset to extract insights and address key issues in predicting hourly passengers per bus stop.
- Secured a top-tier competitive ranking (3rd place) by engineering a high-precision XGBoost model that delivered industry-leading predictive accuracy, setting a new performance benchmark.

### 2nd Place Machine Learning Competition - Airlangga University

Sept. 2025

- Achieved 2nd place out of 200 teams in a Machine Learning Competition at Airlangga University by analyzing prison verdict documents using NLP.
- Engineered a sentence complexity classifier to streamline content processing, enabling 20% faster document handling and more efficient resource allocation for high-volume text analytics.

### 6th Place Data Mining Competition - Telkom University

Nov. 2025

- Achieved 6th place out of 103 teams by developing an end-to-end data analysis pipeline for undersea mammal classification.
- Engineered the WhaleNet computer vision system to solve the manual bottleneck in marine image analysis, achieving over 95% identification accuracy and reducing data-to-insight timelines by 40%.

### 7th Place Data Analyst FIND-IT Competition - Gadjah Mada University

Apr. 2025

- Ranked among the top 10 finalists out of 1,181 teams by developing end-to-end data analysis pipelines and predictive classification models.
- Engineered a predictive model using tabular app metadata to classify COPPA compliance risk, transforming manual regulatory review into an automated, scalable audit system that mitigates legal exposure.

## Certifications

- Supervised Machine Learning and Unsupervised Machine Learning, DeepLearning.AI And Stanford (92.46)
- Linear Algebra for Machine Learning and Data Science, DeepLearning.AI And Stanford (93.32)
- Introduction to SAP S/4HANA with GBI 4.2, Department of Information Systems Institut Teknologi Sepuluh Nopember (84.83)